

EcoBreeze OGG Simulator Performance Manual

ADS4293 Professional Analytics

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1. Overview of Worksheets

1.1 Supporting Sheets

Read Me

Explanation of the worksheet structure; any important notes or version information users should read before using the model

DashBoard

Visual summary of key performance indicators (KPIs), for example: Net profit by quarter, Sales by region, Cash flow trends, Market Share

Designed for quick monitoring, presentation, and management-level review.

Reference

Mainly the list used for “Data Validation”

1.2 Data Sheets

Used for descriptive statistics and data preparation for predictive analysis.

Data_Price

Data_CashFlows

Source data for cash flow information by period and/or by region, used in both descriptive analysis and predictive models.

Data_Stock

Source data for inventory and stock levels, for both raw material and finished goods

Data_IncomeStatement

IncomeStatement

Data_EmployeeNumber

Source data on employee (mainly the sellers) counts by quarter and by region

Data_Sales

Source data for sales volume by region

1.3 Template Sheets

Template

Qm-ValidationPredict

Input_Qm-Predict

The template is designed for the creator to check references and update the model. Users can copy and paste the template, change the quarter number in the top-left corner, and generate a new sheet for the next quarter.

Which Quarter:	
7	
For:	
Template	
Template	
Prediction	Overtime
Validation	

If new indicators need to be added to the model, the template can also be used to test and update them, ensuring that changes can be applied consistently in future quarters. The **Qm-ValidationPredict** sheet contains all detailed calculation steps leading to the income statement. The workbook has two main user sheets: **Predict**, for decision-making, and **Validation**, for reflection and entering empirical (actual) data. **Input_Qm-Predict** is used for prediction analysis: users enter all decision points, and with different inputs they can quickly see the resulting net profit and break-even units.

These two templates were built and first used in **Quarter 6**. For earlier quarters, the information sheets are hidden and were not retroactively updated. This is because we treat the game as a stochastic process: most of our decisions are based on the current quarter's resources and state, so updating prior quarters would add detail without improving current decision-making.

Descriptive Analysis

Qm-Validation (sheet name with m=quarter number)

There is one sheet for each quarter, detailing the decision for the quarter and the actual outcome.

The sheet section mimic the calculation from the OGG, with the sequence of Operation, Marketing, Human Resource and Finance, with the actual quarter result.

Q7-Validation	Q6-Validation	Q5-Validation	Q4-Validation	Q3-Validation
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1.4 Descriptive Analysis & Predictive Analysis Sheets

Descriptive Analysis

Qm-Validation (sheet name with m=quarter number)

The sheet is one for each quarter, detailing the decision for the quarter and the actual outcome.

The sheet section mimic the calculation from the OGG, with the sequence of Operation, Marketing, Human Resource and Finance, with the actual quarter result.

Q7-Validation	Q6-Validation	Q5-Validation	Q4-Validation	Q3-Validation
---------------	---------------	---------------	---------------	---------------

Predictive Analysis

Input_Qm-Predict(sheet name with m=quarter number)

Sheet where decision variables for the upcoming quarter are entered, focuses mainly on predicting net profit for the upcoming quarter such as: Selling prices, Regional distribution of product, Investment amount, etc

Qm-ValidationPredict(sheet name with m=quarter number)

Prediction sheet where detailed calculations are performed based on the inputs from **Input_Qm-Predict**. The sheet section mimic the calculation from the OGG, with the sequence of Operation, Marketing, Human Resource and Finance, with the predict quarter performance.

1.5 Prescriptive Analyses

After the quarter has been run, the team reviews the results together. The previous quarter's performance and the aims for the next quarter are discussed in a group meeting. A short note summarizing these aims and key points is then recorded either:

- at the top of the **Qm-ValidationPredict** sheet for the upcoming quarter, or
- in a separate general report document.

Our prescriptive analysis does not rely on a separate formal model. Instead, we integrate all current information (financial results, operational indicators, and market feedback), identify key issues, and record them as qualitative guidance for the next round of decisions.

2. How to Use the Spreadsheet (Workflow)

Note: The model is not built for the initial decision and Quarter 1.

Work flow start when the new quarter result is out:

1. Update Data from the Results (Manual Input)

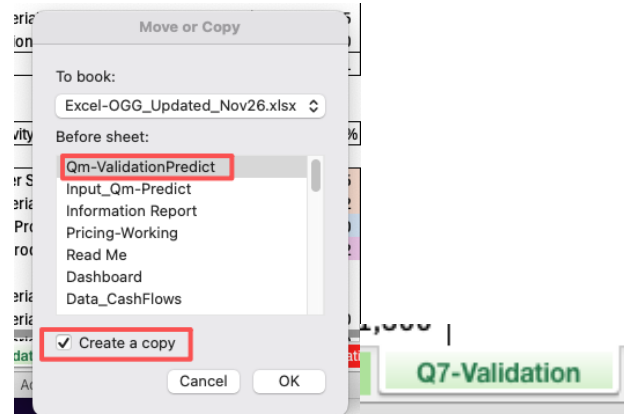
- Manually copy the latest results from the OGG system/game (e.g., sales, income statement, inventory, employee numbers) into the relevant data sheets:

Data_CashFlows
Data_Stock
Data_IncomeStatement
Data_EmployeeNumber
Data_Sales

Data_CashFlows	Data_Stock	Data_IncomeStatement	Data_EmployeeNumber	Data_Sales
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2. Create or Update the Validation Template (Qm-Validation)

- Copy** the Qm-ValidationPredict template for the current quarter m , and name it accordingly (For example, for Quarter 7 validation, sheet name Q7-Validation)



- Choose the quarter number and template type (Validation)

Which Quarter:	7
For:	Validation
Overtime?	Overtime

From last Quarter
Decision Point
Prediction

1. Operation

Q() summary/reflection

- Check if the income statement based on our own calculation matches with the income statement given by OGG.

(There might be some discrepancy on COGS and Freight Cost, which need to be manually adjusted on the sheet)

Our income statement based on the predicted sales, not the planned sale			
Sales Income	2,792,036.00		
COGS	1,033,124.62	Not Match	1033124.62
Gross Profit and Loss	1,758,911.38		
Production Fixed Cost	560,000.00	FC	
Salaries	608,400.00	FC	
Training Investments	70,000.00	FC	
Fringe Benefits Investments	47,000.00	DL	
CSR. Investments	10,000.00	FC	
Profit Sharing Payment	0.00		
Research & Development	135,000.00	FC	
A&P Investments	319,567.59	FC	
Raw Material Storage Costs	20,020.85	FC	
Information Report Purchase	40,000.00	FC	
Freight Costs	38,776.00	Need Input Manually	38776
Loss & Damage	0.00		
Loan Interest	0.00		
Factoring Interest	0.00		
Decision Fine	0.00		
PROFIT & LOSS STATEMENT EXPENSES	1,848,764.44		
(+) Income savings	0		
(+) Bonus Value	0		
Pretax Income	-89,853.06		
(-) IR	0		
NET PROFIT / LOSS	-89,853.06		
PREVIOUS QUARTER BALANCE	-266,518.04		
NET PROFIT / LOSS	-89,853.06		
Fixed sale variation	0		
PROFIT AND LOSS STATEMENT RESULTS	-356,371.10		

3. Complete the Validation and Reflection (Prescriptive Thinking)

- After running the model (“pressing the button”), discuss the results in an in-person or online group meeting and summarize:
- Highlight issues and key points
- Performance evaluation
- Insights from the discussion
- Suggestions and key aims for improvement in the next quarter
- Record these key aims and notes at the top of the **Qm-Predict** sheet for the next quarter and in the separate general report.

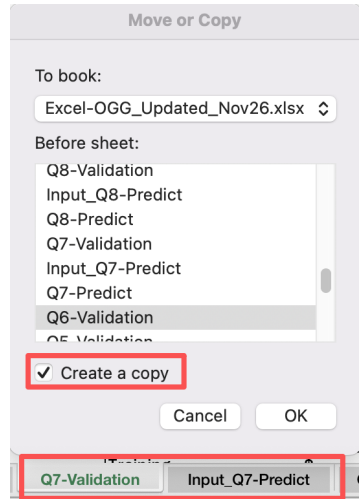
Which Quarter:
7
For:
Validation
Overtime?
Overtime
From last Quarter
Decision Point
Prediction

Q() summary/reflection

1. Operation

4. Prepare the Prediction Templates for the Next Quarter (Predictive Analysis)

- Copy or reset the **Input_Qm-Predict** and **Qm-Predict** sheets for quarter $m+1$, and name it accordingly (For example, for Quarter 7 decision, sheet name **Input_Q7-Predict** and **Q7-Predict**)



- Enter new decision variables in **Input_Qm-Predict** (prices, production, regional allocation, etc.). Only the cells highlighted in red and blue require an input; all the non-highlighted cells have their own function and will automate a result accordingly

Preparation for next quarter prediction				1. Validate data from previous quarter (By QingQing)
☒				2. Update Data_Sheets(Need cash flow update By Seb)
☐				3. Create new prediction sheet for new quarter+ Validation (
☐				4. Choose the Number of current quarter in B6
Quarter		11		
Strategy		Maximum Capacity		Note: Conservative strategy ha
Working Hour		Overtime		
SalePrice				Predicted Sale
	1	\$ 2,899		1,332
	2	\$ 2,399		BreakEven unit
	3	\$ 2,899		1145
	4	\$ 2,899		All Good
NET PROFIT / LOSS		442,087		
PROFIT AND LOSS STATEMENT RESULTS		1,280,875		
Previous Quarter balance		838,788		DO NOT CHANGE THIS CELL!!
Loan interest		-		Given
Climatizer Stock		427		DO NOT CHANGE THIS CELL!!
Raw Material Available		1,992		DO NOT CHANGE THIS CELL!!
Maximum Sell unit		1,925		
Plan to Sell		1,925		# Prediction based on the dema
		1,925	Proportion	# Capped by the maximum sell u
	1	577	0.3	# Plan to have more market shai
	2	327	0.17	# Focus
	3	57	0.03	
	4	962	0.5	# Leader
Predicted Sale		1,332	Proportion	# Based on the predicted conver
	1	403	0.7	
	2	228	0.7	
	3	28	0.5	
	4	673	0.7	

- Quick preview of the Quarter performance

NET PROFIT / LOSS	442,087
PROFIT AND LOSS STATEMENT RESULTS	1,280,875

- There is a small box that provides a quick overview of the break-even units based on the current predicted performance. With fixed costs held constant, the break-even quantity mainly depends on the variable cost per unit. When the predicted sales volume is lower than the break-even units, a warning message (“Need adjustment”) appears. To prevent an anticipated loss, we can then adjust the decision inputs (e.g., price, volume, or cost drivers) and re-evaluate the outcome.

Preparation for next quarter prediction	
☒	1. Validate data from previous quarter
☐	2. Update Data_Sheets
☐	3. Create new prediction sheet for new quarter+ Validation
☐	4. Choose the Number of current quarter in B6
Quarter	11
Strategy	Maximum Capacity
Working Hour	Overtime
SalePrice	
1 \$	2,899
2 \$	2,399
3 \$	2,899
4 \$	2,899
NET PROFIT / LOSS	442,087
PROFIT AND LOSS STATEMENT RESULTS	1,280,875

Note: Conservative strategy have a cons

Predicted Sale	766
BreakEven unit	1145
Need Adjustment	

- Review the detailed predicted results in **Qm-Predict**

5. Finalize Decisions and Input into OGG

6. Repeat the Cycle

- Once the next quarter’s actual results are available, we return to Step 1, update the data, and repeat the process.